

1. Overview

Thank you for purchasing the HPCU parallel control module.

The HPCU module can be used on MD880 series and HE series AC drive to support up to ten paralleled modules. Communicating with the HCU module and power module through optical fiber cables, the HPCU module acts as an intermediate device between them. That is, the HPCU module receives IGBT drive and control signals sent from the HCU and transmits such signals to each power module. Meanwhile, the HPCU module uploads the current, voltage, and status data sent from the power module to the HCU.

 Features

- Controls the synchronization and current sharing of all paralleled power modules
- Depending on system requirements, HPCU-40 (four paralleled modules supported), HPCU-60 (six paralleled modules supported), or HPCU-A0 (ten paralleled modules supported) can be selected.
- Communicates with the HCU module and paralleled modules through optical fibers to ensure reliable system operation
- Equipped with SD card to save fault data
- Equipped with real-time clock for timestamping fault events
- Installed through the guide rail for flexible installation

 Manual obtaining

The information in this user guide is subject to change without notice. You

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can acquire the latest document version from your product distributor.

2. Product Information

◆ Nameplate and model

HPCU - 40

| Label | Product series          |
|-------|-------------------------|
| HPCU  | Parallel Control Module |

Nameplate

Model: HPCU-40

Rated input: DC24V 0.5A

S/N: XXXXXXXXXXXXXXX



| Product type             |
|--------------------------|
| 40 4 paralleled modules  |
| 60 6 paralleled modules  |
| A0 10 paralleled modules |

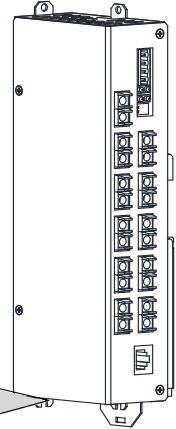


Figure1 Nameplate of the HPCU module

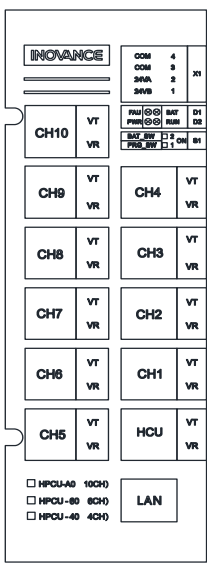
◆ Model

| Model   | Name                          | Description                                       |
|---------|-------------------------------|---------------------------------------------------|
| HPCU-40 | Four Parallel Control Modules | Supports up to four modules installed in parallel |
| HPCU-60 | Six Parallel Control Modules  | Supports up to six modules installed in parallel  |
| HPCU-A0 | Ten parallel Control Modules  | Supports up to ten modules installed in parallel  |

◆ Specifications

| Item                | Description                   |
|---------------------|-------------------------------|
| Input voltage       | 24.0 V±10% at 0.5 A           |
| Ambient temperature | -10℃ to 55℃                   |
| Storage temperature | -40℃ to 70℃                   |
| Relative humidity   | 5% to 95% RH (non-condensing) |
| Ambient conditions  | Free from corrosive gases     |
| Installation method | Guide rail installation       |
| IP rating           | IP20                          |
| Heat dissipation    | Natural ventilation           |

◆ Terminal description



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|      |   |    |
|------|---|----|
| COM  | 4 | X1 |
| 24V  | 2 |    |
| 24VB | 1 |    |

Model: HPCU-40

Rated input: DC24V 0.5A

S/N: XXXXXXXXXXXXXXX



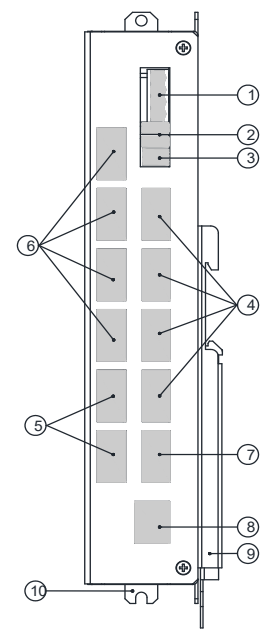
|      |    |     |    |
|------|----|-----|----|
| CH10 | VT | CH4 | VT |
|      | VR |     | VR |
| CH9  | VT | CH3 | VT |
|      | VR |     | VR |
| CH8  | VT | CH2 | VT |
|      | VR |     | VR |
| CH7  | VT | CH1 | VT |
|      | VR |     | VR |
| CH6  | VT | HCU | VT |
|      | VR |     | VR |
| CH5  | VT |     | VT |
|      | VR |     | VR |

☐ HPCU-A0 10030

☐ HPCU-60 8030

☐ HPCU-40 4030

LAN



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Figure2 HPCU module layout

Table1 Description of the HPCU module layout

| No. | Name                      | Terminal    | Note                                   | Description                                                                                                                                                                             |
|-----|---------------------------|-------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ①   | Power supply X1           | COM 4       | 24 V power supply ground               | External power supply of HPCU                                                                                                                                                           |
|     |                           | COM 3       | 24 V power supply ground               |                                                                                                                                                                                         |
|     |                           | 24VA 2      | 24 V power supply +                    | X1 terminal: 5.08 mm clearance, 4-pin black pluggable terminal                                                                                                                          |
|     |                           | 24VB 1      | 24 V power supply +                    | Twisted pair                                                                                                                                                                            |
| ②   | LED indicator             | PWR         | Steady green                           | The HPCU power supply is normal.                                                                                                                                                        |
|     |                           |             | Off                                    | The HPCU module is not energized or the power supply has failed.                                                                                                                        |
|     |                           | RUN         | Steady green                           | The power module is running.                                                                                                                                                            |
|     |                           |             | Off                                    | The power module is stopped.                                                                                                                                                            |
|     |                           | FAU         | Steady red                             | System fault                                                                                                                                                                            |
|     |                           |             | Off                                    | The system is normal.                                                                                                                                                                   |
|     |                           | BAT         | Reserved                               | /                                                                                                                                                                                       |
|     |                           |             | Reserved                               | /                                                                                                                                                                                       |
| ③   | DIP switch S1             | BAT_SW      | Reserved                               | /                                                                                                                                                                                       |
|     |                           | PRG_SW      | Reserved                               | /                                                                                                                                                                                       |
| ④   |                           | CH1 to CH4  | Parallel module channel 1 – channel 4  | VT: 50 M, transmitting optical fiber communication signals<br>VR: 50 M, receiving optical fiber communication signals                                                                   |
|     |                           | CH5 to CH6  | Parallel module channel 5 – channel 6  |                                                                                                                                                                                         |
| ⑤   |                           | CH7 to CH10 | Parallel module channel 7 – channel 10 | Optical fiber type: plastic optical fiber (POF)<br>Channels available for HPCU-40: CH1 – CH4<br>Channels available for HPCU-60: CH1 – CH6<br>Channels available for HPCU-A0: CH1 – CH10 |
|     |                           |             |                                        |                                                                                                                                                                                         |
| ⑥   | Optical fiber transceiver | CH7 to CH10 | Parallel module channel 7 – channel 10 | Optical fiber type: plastic optical fiber (POF)<br>Channels available for HPCU-40: CH1 – CH4<br>Channels available for HPCU-60: CH1 – CH6<br>Channels available for HPCU-A0: CH1 – CH10 |
|     |                           |             |                                        |                                                                                                                                                                                         |
| ⑦   |                           | HCU         | Communication between HPCU and HCU     | Channels available for HPCU-40: CH1 – CH4<br>Channels available for HPCU-60: CH1 – CH6<br>Channels available for HPCU-A0: CH1 – CH10                                                    |
|     |                           |             |                                        |                                                                                                                                                                                         |
| ⑧   | /                         | LAN         | Commissioning terminal                 | Connection with PC                                                                                                                                                                      |
| ⑨   | Guide rail groove         | /           | For guide rail installation            | /                                                                                                                                                                                       |
| ⑩   | Mounting hole             | /           | 4 holes for securing installation      | /                                                                                                                                                                                       |

3. Installation Instructions

◆ Installation precautions

| Item | Notes                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------|
| 1    | Before installation or removal, ensure that the HPCU module is in the power-off state to prevent damage to it. |
| 2    | Protect the HPCU module from falling or impact to avoid damage to it.                                          |
| 3    | Do not disassemble the HPCU module; otherwise, the module may be damaged.                                      |
| 4    | Tighten the screws according to the required torque to avoid damage or loose fastening.                        |

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◆ Tightening torque

The screws must be installed according to the following tightening torque.

Table2 Mechanical connection

| Screw | Tightening Torque |
|-------|-------------------|
| M3    | 1.2 N · m         |
| M4    | 2.9 N · m         |

Table3 Electrical connection

| Screw | Tightening torque |
|-------|-------------------|
| M3    | 0.55 N · m        |

◆ Dimensions

Figure3 Product dimensions (in mm)

◆ Installation clearance requirements

To install the HPCU module, certain space must be reserved as shown below.

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NOTE

◆ Since the optical interfaces and power terminals are on the right side of the HPCU module, the minimum clearance on the right side is equal to or over than 100 mm .

◆ Installation procedure

The HPCU module can be installed through the guide rail or the mounting hole. Installation through the guide rail is recommended and introduced in this user guide.

Figure5 Dimensions of the guide rail (in mm)

Step 1: Tilt the HPCU module and place the upper part of the module's guide rail groove on the guide rail. Press the lower part of the guide rail groove in the direction of the arrow to securely lock the groove (part ①) on the guide rail (part ②).

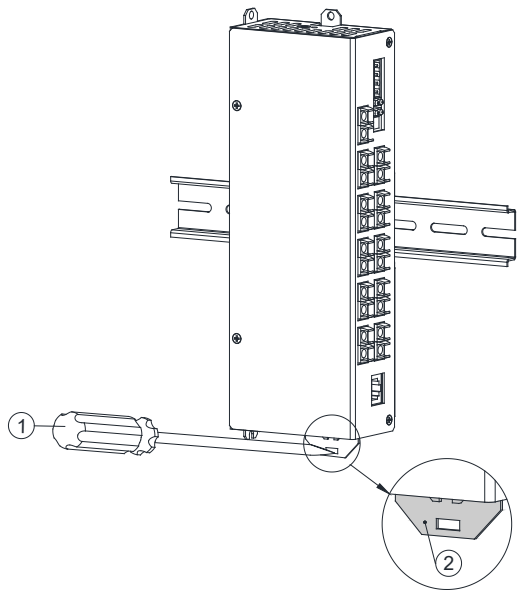
Step 2: The guide rail groove is successfully installed as the following figure.

Step 3: Connect the 24 VDC external power supply to the X1 terminal and the optical fiber cables to the corresponding interfaces to complete the installation.

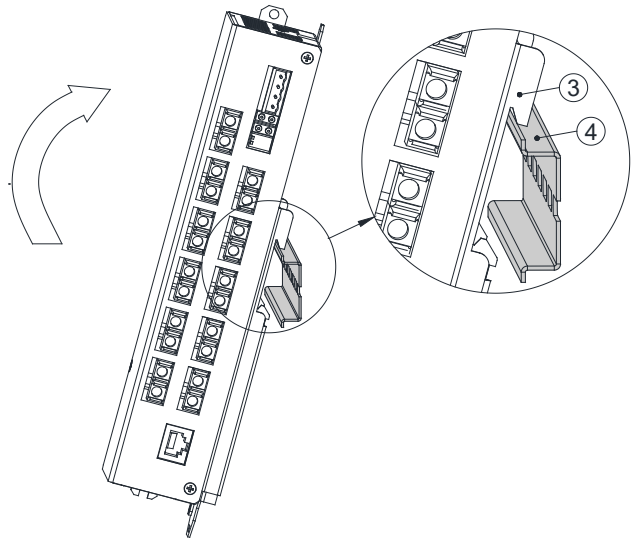
◆ Removal

Step 1: Unplug all cables connected to the HPCU module.

Step 2: Press down with a straight screwdriver ① to release the guide rail lock ②.



Step 3: Rotate the module in the direction of the arrow, and remove the guide rail groove ③ completely from the guide rail ④.



## 4. Electrical Connection

### ◆ Electrical connection diagram

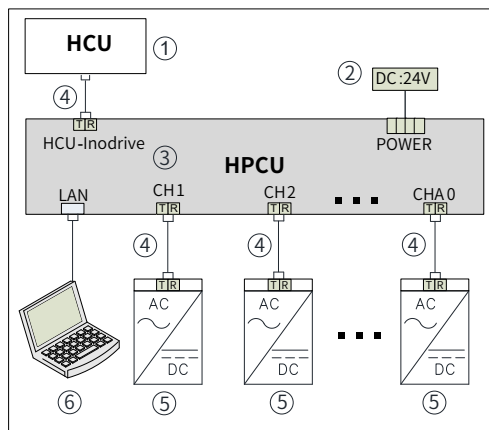


Figure6 Electrical connection of the HPCU module

| No. | Name                         |
|-----|------------------------------|
| ①   | HCU Control Module           |
| ②   | 24 VDC external power supply |
| ③   | HPCU Parallel Control Module |

|   |                             |
|---|-----------------------------|
| ④ | Optical fiber communication |
| ⑤ | Power module                |
| ⑥ | Computer                    |

### ◆ Optical fiber requirements

To ensure reliable operation, it is recommended to order optical fiber cables manufactured by Inovance. The specifications and models are as follows:

Table4 Standard fiber optic cable selection

| Code     | Name                                    | Specifications |
|----------|-----------------------------------------|----------------|
| 01620327 | MD880 fiber optic cable - 1.5 m (ROHS)  | 1.5 m/1 PCS    |
| 01620328 | MD880 fiber optic cable - 3.0 m (ROHS)  | 3.0 m/1 PCS    |
| 01620329 | MD880 fiber optic cable - 5.0 m (ROHS)  | 5.0 m/1 PCS    |
| 01620330 | MD880 fiber optic cable - 7.0 m (ROHS)  | 7.0 m/1 PCS    |
| 01620331 | MD880 fiber optic cable - 10.0 m (ROHS) | 10.0 m/1 PCS   |
| 01620332 | MD880 fiber optic cable - 15.0 m (ROHS) | 15.0 m/1 PCS   |
| 01620333 | MD880 fiber optic cable - 20.0 m (ROHS) | 20.0 m/1 PCS   |



NOTE

Each product code contains only one optical fiber. For normal use, purchase it in pairs.

Table5 Optical fiber specifications

| Item                           | Specifications              | Item                          | Specifications |
|--------------------------------|-----------------------------|-------------------------------|----------------|
| Material                       | Plastic optical fiber (POF) | Storage temperature           | -55℃ to +85℃   |
| Maximum short-term tension     | 50 N                        | Installation temperature      | -20℃ to +70℃   |
| Minimum short-term bend radius | 25 mm                       | Minimum long-term bend radius | 35 mm          |
| Maximum long-term tension load | 1 N                         | Maximum bend times            | 1000           |
| Maximum connection distance    | 30 m                        | /                             | /              |



Figure7 Appearance of the optical fiber



NOTE

- ◆ When plugging and unplugging the optical fiber cable, pull the optical fiber connector, not the cable itself. Since the optical fiber cable is extremely sensitive to dirt, do not touch the end of the cable with bare hands.
- ◆ To ensure the successful fiber assembly, connect the gray fiber connector to the gray fiber transmission interface and connect the black fiber connector to the black fiber receiving interface.

## 5. Parameter Configuration

◆ Configure the HPCU module through the HCU control module  
Configuration flowchart

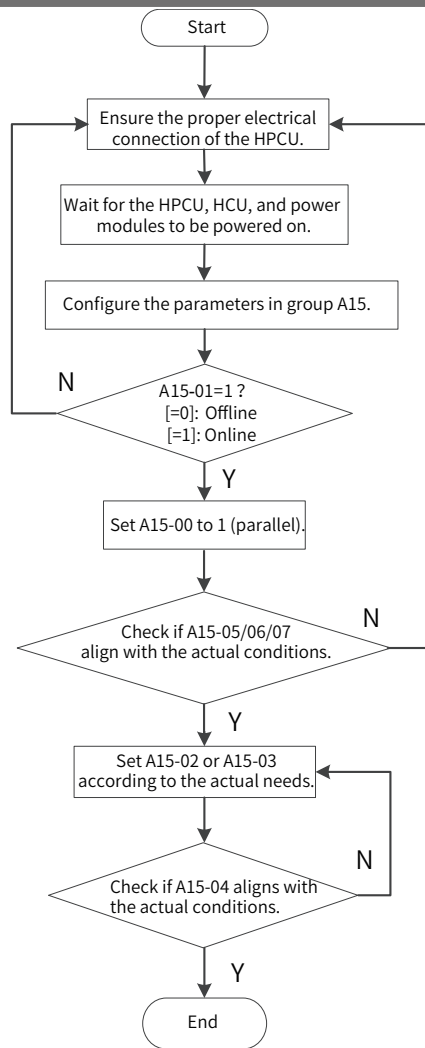


Figure8 HPCU parameter configuration flow

Table6 Parameter description

| Item | Param. | Parameter name                      | Note                                                                                                                             |
|------|--------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1    | A15-00 | Parallel function selection         | 0: Single unit<br>1: Parallel                                                                                                    |
| 2    | A15-01 | Online indicator of the HPCU module | 0: Offline<br>1: Online<br>Read-only.                                                                                            |
| 3    | A15-02 | Parallel module enable selection 1  | The first digit to the last digit of 00000 corresponds to module 5 to 1, respectively.<br>0: Disable<br>1: Enable                |
| 4    | A15-03 | Parallel module enable selection 2  | The first digit to the last digit of 00000 corresponds to module 10 to 6, respectively.<br>0: Disable<br>1: Enable<br>Read-only. |
| 5    | A15-04 | Number of enabled parallel modules  | After the A15-02 and A15-03 are set, the number of enabled parallel modules are automatically calculated.                        |
| 6    | A15-05 | Parallel module online display 1    | Read-only.                                                                                                                       |
| 7    | A15-06 | Parallel module online display 2    | Read-only.                                                                                                                       |
| 8    | A15-07 | Number of online parallel modules   | 0 to 10                                                                                                                          |

## INOVANCE Warranty Agreement

Inovance provides an 18-month free warranty (subject to information indicated by the barcode on the equipment) to the equipment itself from the date of manufacturing for the failure or damage under normal use conditions.

Within the warranty period, maintenance will be charged for the damages:

- 1) improper use, disassembly, repair, or modification
- 2) Damage caused by fire, flood, abnormal voltage, other disasters, and secondary disasters;
- 3) employees dropping or improperly transporting the product
- 4) Operations not following the user instructions
- 5) Faults and damages caused by external factors, such as peripheral devices.

If there is any failure or damage to the product, correctly fill out the Product Warranty Card.

Maintenance is charged according to the latest Maintenance Price List of Inovance.

The Product Warranty Card is issued only once and cannot be re-issued. Keep the card and present it to the maintenance personnel when seeking maintenance.

If there is any problem during the service, contact Inovance or Inovance agents.

You are assumed to agree on terms and conditions of this warranty agreement by purchase of the product. This agreement shall be interpreted by Suzhou Inovance Technology Co., Ltd.

## INOVANCE Product Warranty Card

|                      |                                  |                 |
|----------------------|----------------------------------|-----------------|
| Customer Information | Company address:                 |                 |
|                      | Company name:                    | Contact person: |
|                      | P.C.:                            | Tel.:           |
| Product Information  | Product model:                   |                 |
|                      | Product barcode (attached here): |                 |
|                      | Agent name:                      |                 |
| Fault Information    | (Service date and description):  |                 |
|                      | Serviced by:                     |                 |